SPECIFICATION FOR APPROVAL

DESCRIPTION	**High Current Inductor**
PART NUMBER	**EMA5230S Series**
CUSTOMER	
DATE	
AEC-Q200 Qualified	

Index

RoHS Compliant

Revision	Orignated by	Reason for Change	Approved by
2015.07.03	J.G. Lee	First release	B.G. Shim
2017.12.04	L.Kim	Add 5R6	D.G.Yun

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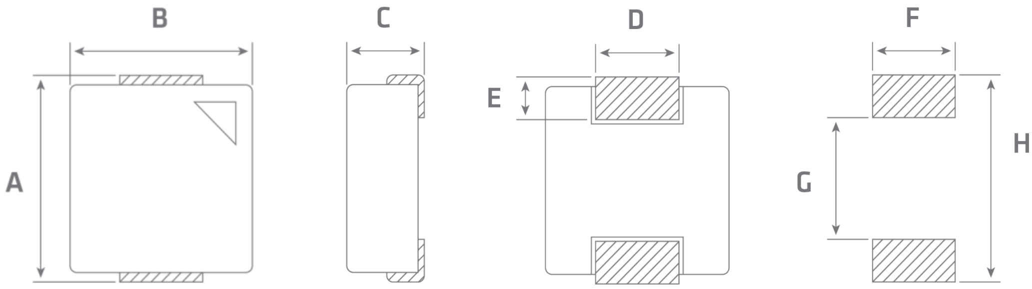
1 Dimension / Quantity



TYPE	EMA5230S	
A	5.4 ± 0.3	mm
B	5.2 ± 0.2	mm
C	3.0 max	mm
D	2.2 ± 0.3	mm
E	1.2 ± 0.3	mm
F	2.5	mm
G	2.2	mm
H	6.0	mm
Quantity	2,000	pcs

2 Electrical Specifications

Part Number	Inductance [uH]	DC Resistance [typ., mΩ]	DC Resistance [max., mΩ]	Saturation Rated Current [max.,A] [1]	Temperature Rise Current [typ.,A] [2]
EMA5230S - R20M	0.20 ± 20%	3.5	3.9	14.5	14.0
EMA5230S - R47M	0.47 ± 20%	7.4	8.5	12.0	11.0
EMA5230S - R68M	0.68 ± 20%	11.0	12.0	11.5	9.0
EMA5230S - 1R0M	1.0 ± 20%	13.0	14.0	11.0	8.5
EMA5230S - 1R2M	1.2 ± 20%	15.0	16.0	11.0	8.5
EMA5230S - 1R5M	1.5 ± 20%	20.0	25.0	8.5	8.2
EMA5230S - 2R2M	2.2 ± 20%	25.0	29.0	7.5	7.0
EMA5230S - 3R3M	3.3 ± 20%	32.0	38.0	6.0	5.5
EMA5230S - 4R7M	4.7 ± 20%	50.0	60.0	5.0	4.5
EMA5230S - 6R8M	6.8 ± 20%	75.0	90.0	4.0	3.5
EMA5230S - 100M	10.0 ± 20%	110.0	125.0	3.5	3.2

- All Test Data is Referenced to 25°C Ambient
- Measuring Condition : 100kHz, 100mV
- Tolerance of Inductance : ± 20%
- [1] Saturation Rated Current : 30% lower than its initial value
- [2] Temperature Rise Current : ΔT = 40°C
  (Temperature of components should be checked in the end application)
- Operating Temperature : -55~125°C

3 Part Number Information

EM A 5230 S □□ - 4R7 M

① ② ③ ④ ⑤⑥ ⑦ ⑧

❶ Product Series

Mark	Series
EM	Metal Molding Design

❷ Core Type

Mark	Core Type
A	Metal Alloy
B	-
C	Carbonyl

❹ Shape

Mark	Shape
T	Triangle
S	Square
R	Rectangle
P	Pentagon
H	Hexagon
O	Octagonal
C	Circle

❼ Inductance Value

Mark (Ex.)	Induction Value [uH]
R47	0.47
4R7	4.7
100	10

❸ Dimension

Mark	Dimension (L*W*H) [mm]
4012	4.0 * 4.3 * 1.2
4020	4.0 * 4.3 * 2.0
5218	5.2 * 5.4 * 1.8
5230	5.2 * 5.4 * 3.0
6515	6.5 * 7.1 * 1.5
6518	6.5 * 7.1 * 1.8
6524	6.5 * 7.1 * 2.4
6530	6.5 * 7.1 * 3.0
6540	6.5 * 7.1 * 4.0
10040	10.0 * 11.5 * 4.0
13050	12.7 * 13.2 * 5.5
17075	17.0 * 17.9 * 7.5

❺❻ Addition

Mark	Addition
*	Reserve
*	Reserve

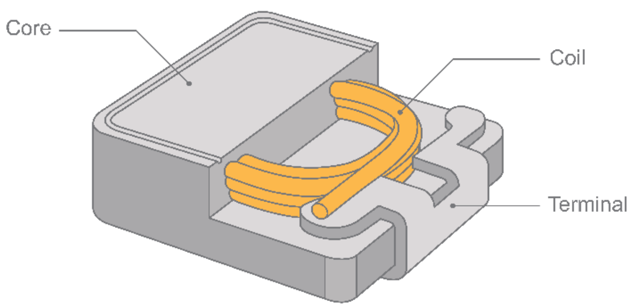
❽ Inductance Toleance

Mark	Tolerance [%]
K	±10
M	±20
N	±30

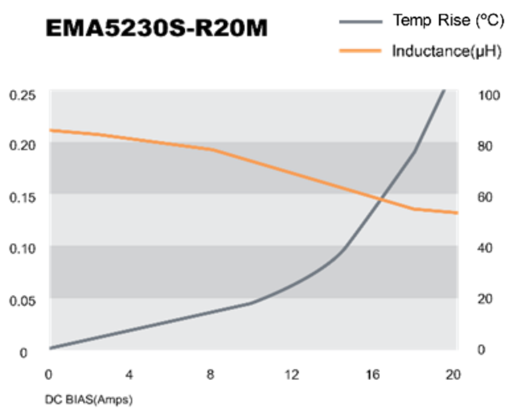
4 Product Profile

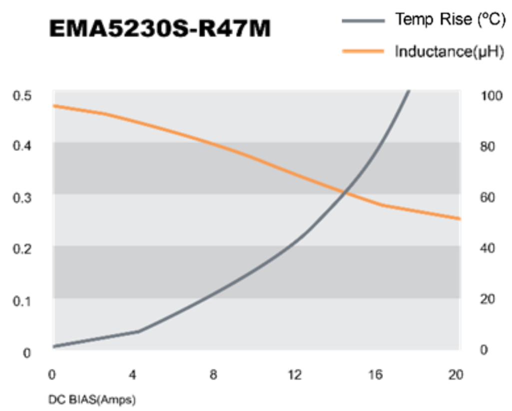
el3 COIL

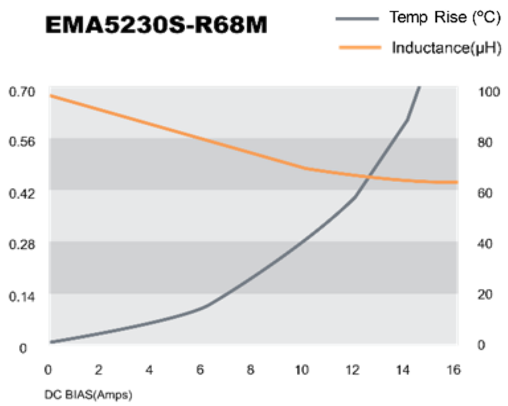
SMD High Current Inductor EM Series

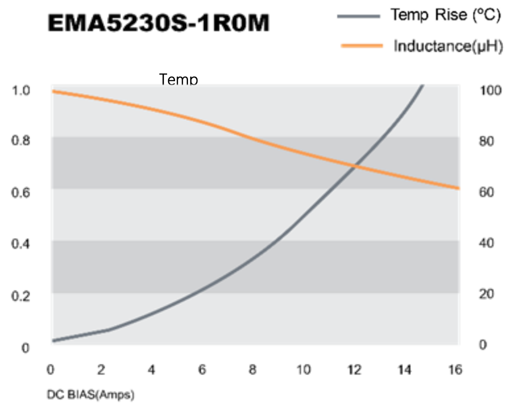


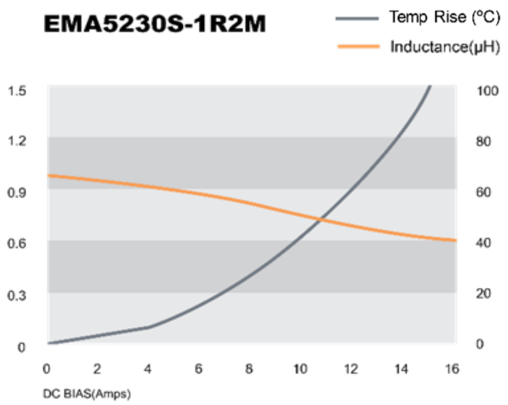
5 Performance Graphs

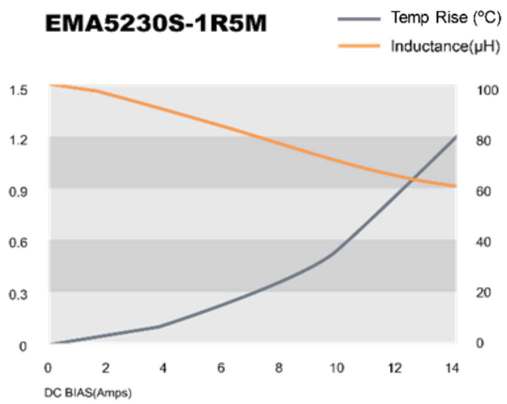












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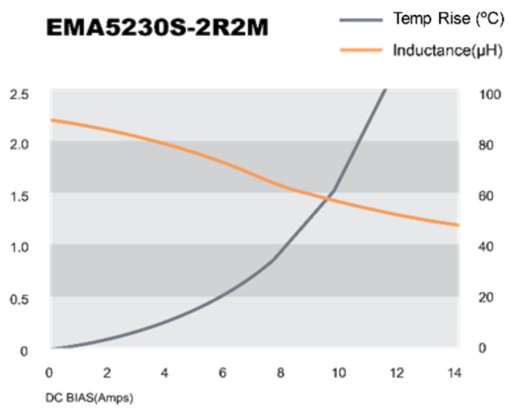
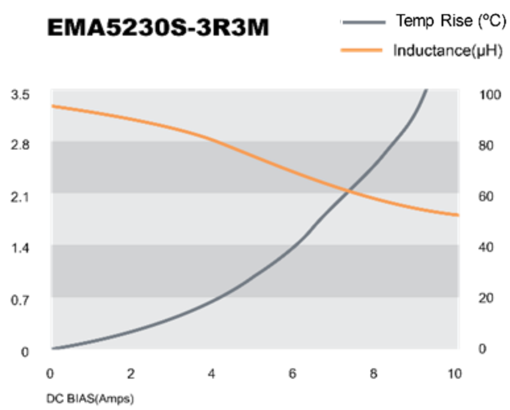
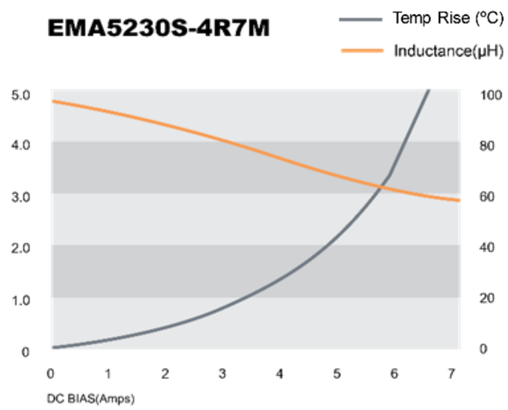
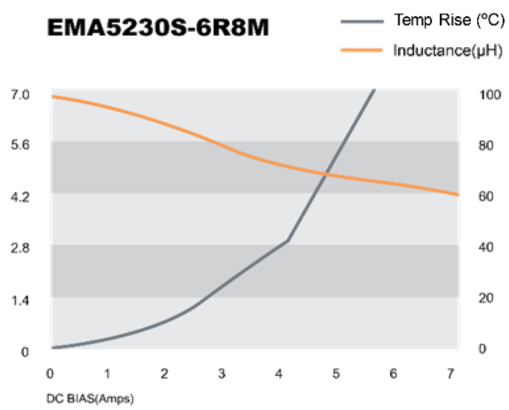
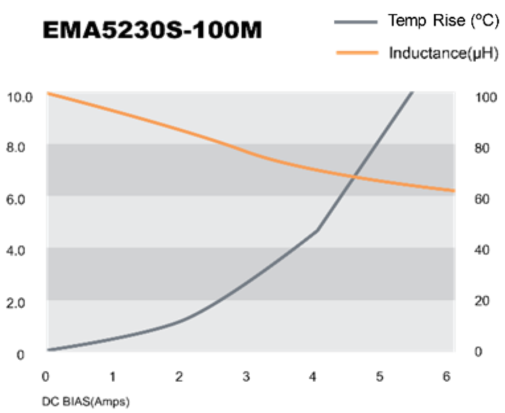










6 Electrical Performance Test

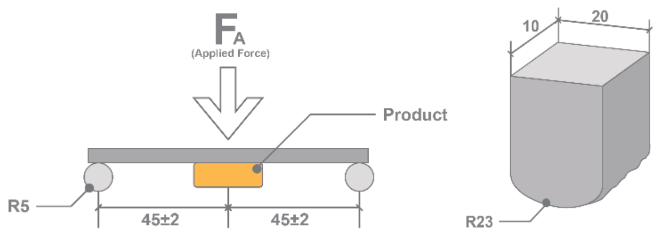
Item	Specification and Requirement	Test Conditions and Method
Inductance	See 3page Technical Data	Agilent 4284A PRECISION LCR METER or equivalent
DC Resistance		HIOKI 3540 mΩ HiTESTER or equivalent
Temperature Rise	40ºC Ref	Coil temperature rise measured at normal operating temperature and rated current using resistance change method or thermostatic thermometer

7 Mechanical Characteristics

Substrate Bending

Test Conditions and Method

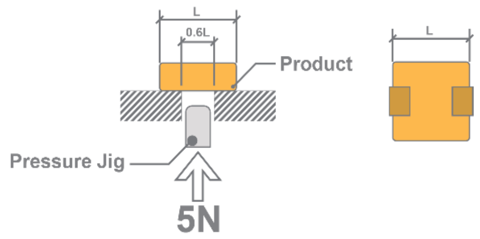
The product shall be soldered onto the printed circuit board in figure and the load is applied until the bending in the arrow direction is not less than 2mm (keep time 4±1seconds)
The test shall be performed totally 3 times.
The PCB is LPF 4027-B standard board.
Requirement :There shall be no mechanical damage ΔL/Lo≤±5%



Product Strength

Test Conditions and Method

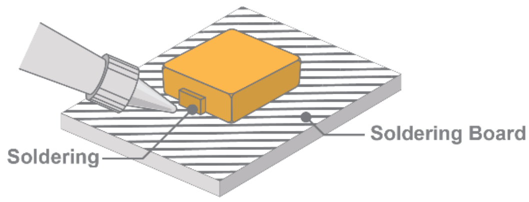
The center of product shall be loaded with a pressure jig (5±0.,5N) during 10±1 seconds.
Requirement : There shall be no mechanical damage ΔL/Lo≤±5%



Terminal Pull Strengh

Test Conditions and Method

The product shall be soldered onto the PCB and as below pictures force a product to 5±0.5N in 10±1 seconds.
Requirement : There shall be no mechanical damage ΔL/Lo≤±5%



Vibration

Test Conditions and Method

Being soldered on PCB board, a product shall be had a vibration test on an amplitude of 0.7mm P-P or 5G, vibration frequency shall be 10 to 500Hz, the sweeping shall be at 1 octave/min.
and these tests will be kept on each direction (X,Y,Z) for 2 hours. (Total 6 hours)
Requirement : There shall be no mechanical damage ΔL/Lo≤±5%

Drop

Test Conditions and Method

There shall be dropped on a concrete floor form 1meter height. The test shall be performed totally 3times.
At first, drop test will be performed to measure electrical characteristics once.
Then, inspection for appearance damage after 2 times of drop.
Requirement : There shall be no mechanical damage ΔL/Lo≤±5%

8 Environment Characteristics

Temperature Characteristics

Specification and Requirement	Test Conditions and Method
ΔL/Lo(20℃)≤±10%	The test is performed after the product has stabilized in an ambient temperature ranging from -20℃ to +85℃ and the inductance change is calculated based on the value in normal temperature and humidity.

High Temperature Storage

Specification and Requirement	Test Conditions and Method
There shall be no mechanical damage ΔL/Lo≤±5%	There shall be left in a temperature of +85±2℃ for 500±4 hours. Upon completion of the test, the measurement shall be made after the product has been left in a normal temperature and humidity more than 1 hour.

Low Temperature Storage

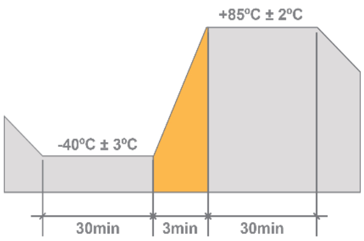
Specification and Requirement	Test Conditions and Method
There shall be no mechanical damage ΔL/Lo≤±5%	There shall be left in a temperature of -40±3℃ for 500±4 hours. Upon completion of the test, the measurement shall be made after the product has been left in a normal temperature and humidity more than 1 hour.

Damp Heat Steady state

Specification and Requirement	Test Conditions and Method
There shall be no mechanical damage ΔL/Lo≤±5%	There shall be left in a temperature of 60±2℃, and humidity of 90~95% for 500±4 hours. With flowing IDC current. Upon completion of the test, the measurement shall be made after the product has been left in a normal temperature and humidity more than 1 hour.

Thermal Shock

Specification and Requirement	Test Conditions and Method
There shall be no mechanical damage ΔL/Lo≤±5%	This test shall be performed upon Completion of 100 cycles in accordance with the condition of 1 cycle in the figure below. Upon completion of the test, the measurement shall be made after the product hasbeen left in a normal temperature and humidity more than 1 hour.



Short time Over load

Specification and Requirement	Test Conditions and Method
There shall be no damage such as smoke or spark	Applied 5 minutes to 1.5 times of DC current

Insulation Resistance

Specification and Requirement	Test Conditions and Method
There shall be no damage Insulation Resistance>1×108Ω	DC 50V voltage shall be applied for 30 sec. between across top surface of the body and the terminal. The insulation resistance shall be more than 1×108Ω.

8 Environment Characteristics

Dielectric With Standing Voltage

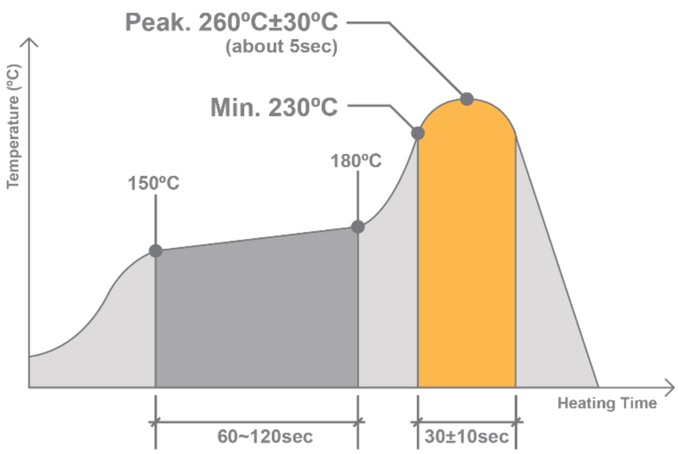
Specification and Requirement	Test Conditions and Method
There shall be no damage	Ac 100V voltage shall be applied for 1 minute across the top surface of the body and the terminal

Solderability Test

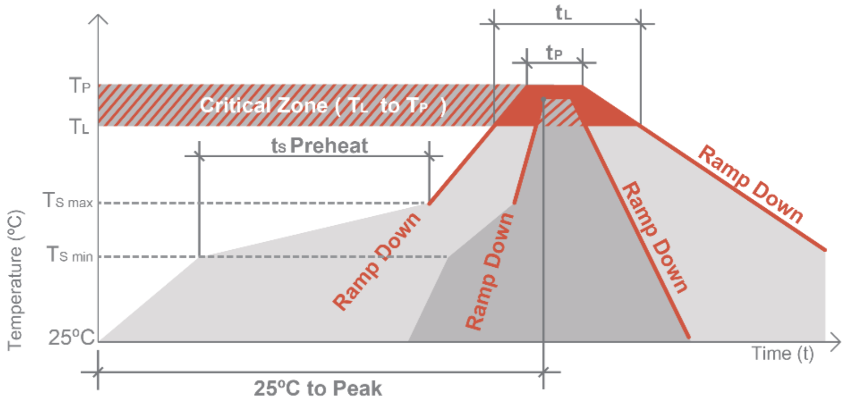
Specification and Requirement	Test Conditions and Method
Over 90% of the electro Sections should be covered with new solder.	Before test : temp. 80℃ humidity 95% 16 hours. **Test** Flux shall be coated over the whole of the product before test, the product shall then be preheated for about 2 minutes in temperature of 150~180℃ and after it has been immersed to a depth 1.0mm below for 3±1 seconds fully in molten solder with a temperature 245±5℃ More than 90% of the terminal electrode sections shall be smoothly covered with new solder when the product is taken out of the solder bath. Remark solder : M705 (SENJU METAL INDUSTRY)

Resistance To Soldering Heat [Reflow Soldering]

Specification and Requirement	Test Conditions and Method
There shall be no mechanical damage ΔL/Lo≤±5%	There shall be passed through the reflow oven with the condition shown in the above profile for twice. The product shall be stored at standard atmospheric conditions for 24 hours, after which the measurement shall be made. Test board : LPF4027-B standard board. Above profiles were measured at the Solder paste part. Solder paste : TLF-204-19A(TAMURA) or equiv.

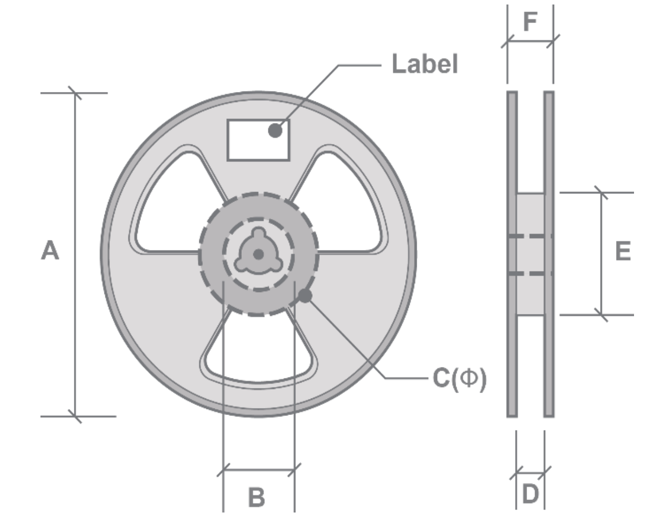


9 Reflow Condition

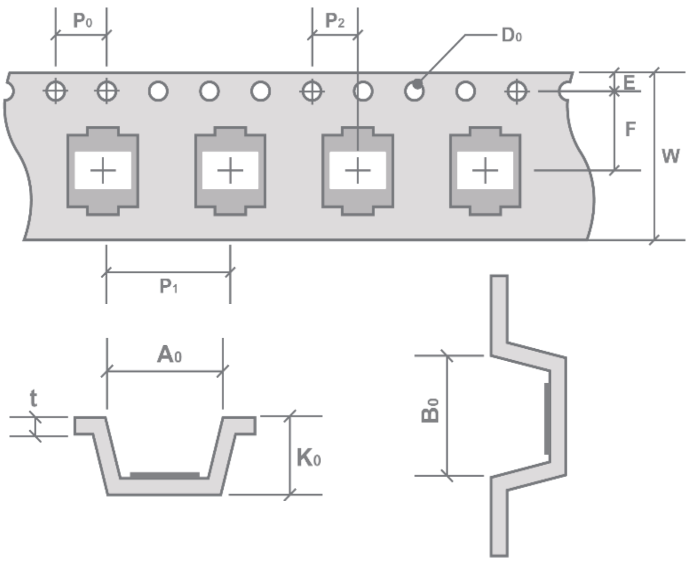


Profile Feature		Sn-Pb Eutectic Assembly		Pb-Free assembly	
		Large Body	Small Body	Large Body	Small Body
Average ramp-up rate (T_L to T_P)		3℃/second max.		3℃/second max.	
Preheat	Temperature Min(T_{smin})	100℃		150℃	
	Temperature Min(T_{smin})	150℃		200℃	
	Time (min to max) (t_s)	60~120 seconds		60~180 seconds	
Ramp-up Rate (T_s max to T_L)		-		3℃/second max.	
Time maintained above	Temperature(T_L)	183℃		217℃	
	Time(t_L)	60-150 seconds		60-150 seconds	
Peak Temperature (T_P)		225+0/-5℃	240+0/-5℃	245+0/-5℃	255+0/-5℃
Time within 5℃ of actual Peak Temperature (t_p)		10~30 sec	10~30 sec	10~30 sec	20~40 sec
Ramp-down Rate (Time 25℃ to Peak Temperature)		6℃/second max.		6℃/second max.	
		6minutes max		8minutes max	

10 Dimension for Carrier Tape & Reel

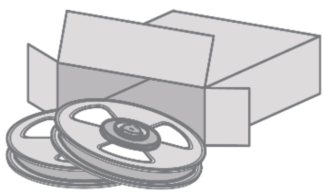


Position	Dimension	
A	330.0±2.0	mm
B	20.0±0.5	mm
C	13.0±0.5	mm
D	12.5±0.2	mm
E	100.0±0.5	mm
F	18.0±1.0	mm



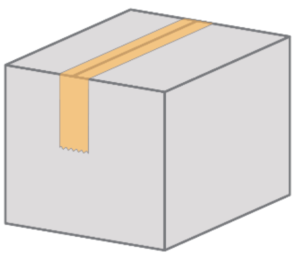
Type	EMA5230S	
Quantity	2,000	pcs
P_0	4.0±0.1	mm
D_0	1.5±0.1	mm
E	1.75	mm
A_0	5.7±0.1	mm
B_0	5.9±0.1	mm
W	12±0.3	mm
P_1	8±0.1	mm
P_2	2±0.1	mm
F	5.5±0.1	mm
K_0	3.6±0.15	mm
t	0.35±0.05	mm

11 Packing Quantity



345mm x 345mm x 55mm(H)

X6 Box =



370mm x 370mm x 380mm(H)

Pack	Quantity
1 Reel	2,000 pcs
1 Middle Carton	2 Reels
1 Big Carton	6 Middle Cartons
	24,000 pcs